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We highly appreciate your commitment and dedication in this endeavour.

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INTRODUCTION

The standard 5 Primary Mathematics Teachers Guide is a document designed to assist teachers to better understand the Standard 5 Primary Mathematics Syllabus. It is designed such that it guides teachers on some activities and materials they can use in addressing some of the syllabus objectives. The Teachers Guide should be used in conjunction with the syllabus. It addresses all modules covered in the Syllabus. This book provides a brief overview for each module. Suggested learning activities and support materials relevant to the activities are based on the specific objectives. Each module has suggested concepts and skills related to the topics being considered. Suggestions where infusion of emerging issues is possible are given in the Guide. The Guide also covers key words to support work in developing mathematical vocabulary effectively. A glossary is provided to explain new, broad and loaded terms used within the Guide.

The information and activities in the Teachers Guide are simply suggestions and should not be treated as prescriptions. Teachers need to consult other reference books for more information in order to develop other activities relevant and accessible to learner's environment. It is also important for the teacher to recap pupils previous knowledge accrued at standard one to four. The activities designed should be learner-centered.

MODULE 1: Numbers and Operations

Module Overview

The module reviews and reinforces what the pupils learned at lower primary. It covers reading and writing numbers up to 10000, the four basic operations, fractions and decimals. It also covers money, which is a manipulative way by which learners can develop number concept within context.

TOPIC: Numbers

Specific Objectives:

1.1.1.1. -1.2.1.8

Content:
Concepts: numerals, more than, less than, equals, odd, even and prime numbers, place value, numbers in expanded form, rounding off
Skills: classifying, reading, writing, and comparing

Support Materials:

Concrete and semi-concrete objects e.g. counters unfix thousand, tens and units' materials, die, flash cards.

Suggested Teaching/Learning Activities:

- Read and write numbers (*finger spell for the deaf*) matching numbers with number names.
  Example: 5302 Five thousand three hundred and two.

  12 Let learners make number patterns from the hundred square chart (they can identify the rules).

  Example:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	57	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

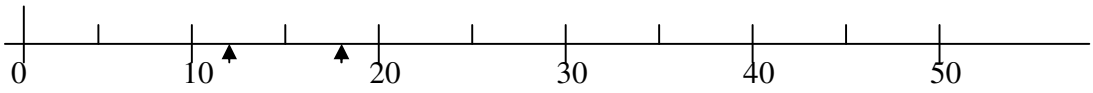
- Use abacus chart to illustrate place value of numbers

TTH	TH	H	T	U
4	3	7	1	2

- Demonstrate expansion of numbers e.g. 4329 written as 4000 + 300 + 20+ 9.

- Show learners how to compare numbers e.g. 300 < 3000. It is important that learners should understand place value in order to compare numbers.

- The number line can be used to elaborate the concept of rounding off as shown in the example.



12 is nearer 10 than 20 hence 12 is rounded off to 10
18 is nearer to 20 than 10 hence 18 is rounded off to 20.

- Guide learners to follow the rounding off rule:
  Numbers less than 5 are rounded down e. g 46**3** is 460 to the nearest ten.
  Numbers from 5 and more are rounded up e. g 46**7** is 470 to the nearest ten.

- Let learners explore various activities in their community that include the use of numbers e.g. police reports on number of people involved in road accidents, shopping activities class/school population, number of people living with HIV/AIDS from clinic or hospital records, herds of cattle reared by farmers, number of loads of sand used to build a certain building, number plates and phone numbers.
  NB* Include community population, elections results, P.M.T.C.T numbers and other current affairs issues

Problem Areas:
Avoid confusing place value and value of digits. There is a need for an example to show place values and value of numbers as shown:

4 3 2 5 has a place value of 4 thousand and the value of the number is four thousands three hundred and twenty five.
Emphasise subtraction from a 3 or 4 digit number with several zeros e.g.

(i) 1000
-721➔ take one from ten and remain with nine.

(ii) For addition, use expanded form e.g. 1 234

$+ 721 \rightarrow 1000 + 200 + 30 + 4$

$700 + 20 + 1$

$1000 + 900 + 50 + 5$

TOPIC: Operations

Specific Objectives: 1.2.1.1 - 1.2.1.19

Content:
Concepts: Estimations, differences, products and quotients, cumulative and associative laws, identity element, factors and multiples, order of operation.
Skills: estimation, addition, subtraction, multiplication and division.

Support Materials:
Concrete and semi-concrete objects.

Suggested Teaching/learning activities:

- Estimation is a fundamental concept that gives learners a reference point to work from. Engage learners in activities that will help them practise the skill of estimation to address objectives 1.2.1.1, 1.2.1.5, 1.2.1.10 and 1.2.1.14.

  Example:

  Addition: $827 + 218 \approx 800 + 200 \approx 1000$ (to the nearest hundred)

  Or $\approx 830 + 220 \approx 1050$ (to the nearest ten)

  Actual $= 1045$

  It might become more understandable if the activity is put in context. For instance, Neo bought to 827 bricks to build her storehouse. She might need 218 more to finish it. Calculate roughly how many bricks she might need altogether to build her storehouse.

  Subtraction: $827 - 218 \approx 800 - 200 \approx 600$

  Or $\approx 830 - 220 \approx 610$

  Actual $= 609$

  Multiplication: $5 \times 39 \approx 5 \times 40 \approx 200$ $\quad$ $12 \times 61 \approx 10 \times 60 \approx 600$

  Division: $48 \div 5 \approx 50 \div 5 \approx 10$

A) Demonstrate the commutative law of addition. It could be in the form of investigation. Exchange order of numbers and let learners find out the result. They could compare the results on the left and right make conclusions.

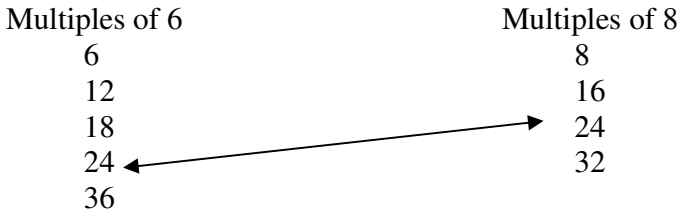
E.g. 75 + 94 = ----- 94 +75 = ------
83 + 16 = ----- 16 + 83 = ------
Hence 75 + 94 = 94 + 75

B) Demonstrate the associative law of addition. You might ask questions like 'does the order of numbers affect the sum?', hence the conclusion that:
72 + (28 + 46) = (72+28) + 46.
NB. *The associative law of addition & multiplication states that when adding or multiplying 3 or more numbers the order does not affect the sum or product.*

- Let learners add zero to several numbers and find the sum. This leads learners to understanding that when adding any number to an identity element, which is zero the number does not change.

Example:
37 + 0 = 37 0 + 19 = 19

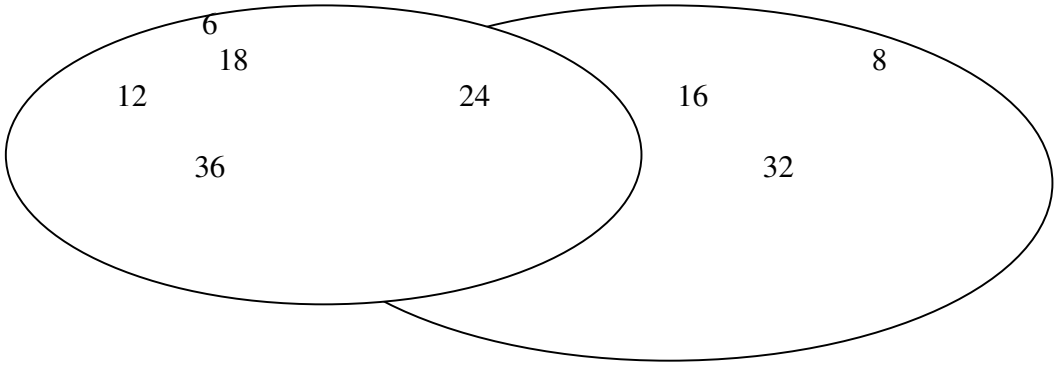
- Let the learners explore multiples of different numbers. Let them compare two sets of multiples e. g
Multiples of 8 less than 40 (8,16,24.32)
Multiples of 6 less than 40 (6,12,18,24,30,32)
You may use one to one correspondences



Alternatively, you may use a Venn Diagram.

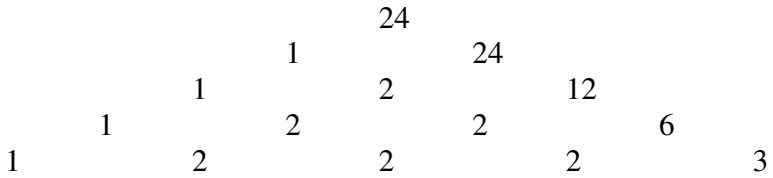
NB: PLEASE DO NOT DWELL MUCH ON SET CONCEPTS.

Venn diagram:

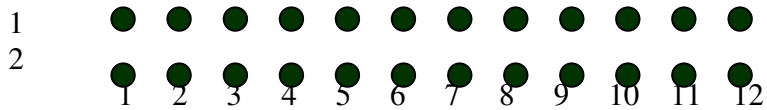


Therefore 24 is the LCM

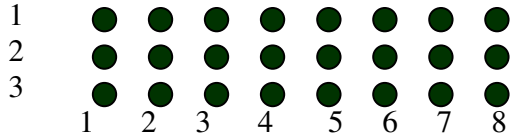
❖ A) Guide learners to use a number pyramid of multiplication to factors of numbers.
e. g. 24



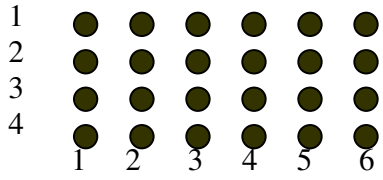
B) Give counters and let them arrange them such that there is no remainders e.g**.**



2 and 12 are factors of 24



3 and 8 are factors of 24



4 and 6 are factors of 24

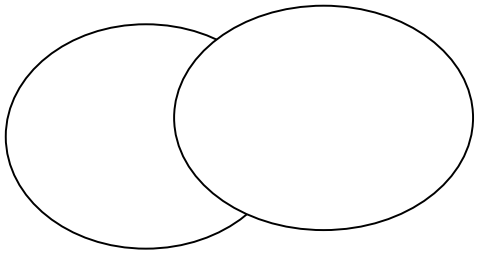
C) Give learners as many numbers as possible to find their factors. Then let them compare factors of two numbers. They may use one to one correspondences to find the Common Factors then choose the Highest Common Factor. E.g.

Factors of 24 (1,2,3,4,5,6,8,12,24)
Factors of 36 (1,2,3,4,6,9,12,18,36)-one to one correspondence.

Factors of 24		**Factors 36**
1	⟷	**1**
2	⟷	**2**
3	⟷	**3**
4	⟷	**4**
6	⟷	**6**
8		9
12	⟷	**12**
24		18
		36

Therefore 12 is the highest common factor
You may use a Venn Diagram

Factors of 24 factors of 36



Venn diagram

12 is the H.C.F of 24 and 36

A) When multiplying a whole number by 10, 100 etc, the zeros in the multiplier are added to the whole number e.g. 15 x 100 = 1500. Hundred has 2 zeros therefore add 2 zeros to 15.

B) Show learners how to use place value to work out multiplication e.g.

TH	H	T	U
	3x10	2x10	5x10
3 ←	2 ←	5 ←	0

325 x 10 = 3250

TTH	TH	H	T	U
		3x100	2x100	5x100
3	2	5	0	0

325 x 100 = 32500

- When we multiply by 10, digits change place value by one column.
- When we multiply by 100, digits change place value by two columns.
- When we multiply 1000, digits change place value by three columns.

❖ Demonstrate that when we multiply a whole number by multiples of 10, zero is put in the unit column as a placeholder.

TH	H	T	U
	3	1	6
	x	2	0
6	3	2	0

NB: When dividing by10, 100. 1000 the reverse of multiplication occurs.

TH	H	T	U
	4 ÷10	2 ÷ 10	0
		4	2

420 ÷ 10 = 42

TH	H	T	U
5	3	0	0
÷ 100	÷ 100		
		5	3

5300 ÷100 = 53

❖ Use the correct order of operations to solve the following three steps number sentences

141 + 127 + 694 =
23 – 11 + 167 =
69 – 6 – 24 =
3 x 4 + 6 =
8 + 24 ÷ 4 =
432 + (119 + 82) =

Engage learners in a discussion to clear some misconceptions or confusion caused by the order of operations. For example, discuss why the answer for 3 x 4 + 6 is 18 but not 30 and why the one for 8 + 24 ÷ 4 is not 8 but 14.

❖ Infuse Problem solving oriented questions in the word problems that you give to learners. Infusion of other issues such as HIV and AIDS is possible here.

TOPIC: FRACTIONS

Objectives: 1.3.1.1 - 1.3.1.13

Contents:
Concepts: *Numerator, denominator, Equivalent fractions, mixed numbers proper and improper fractions, fractions with same and different denominators, fractions, percentages*

Support materials:
Concrete and semi-concrete objects, shapes, fraction chart

Suggested Teaching/learning activities:

Fraction Chart

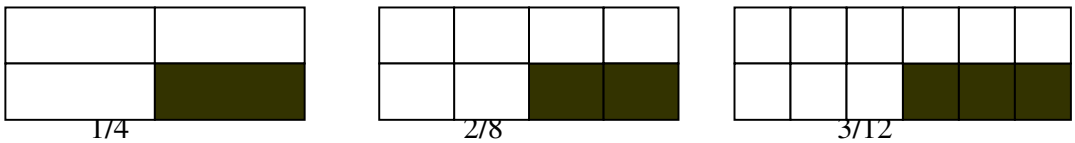
1 whole							
1/2				1/2			
1/4		1/4		1/4		1/4	
1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
1/3			1/3			1/3	
1/6	1/6	1/6		1/6	1/6		1/6

- Use the fraction to compare fractions of different denominators, e. g

  1/8 < 1/6, 3/8 > 1/3

- Show equivalence of fractions

a)



1/4 2/8 3/12

The above shaded sections show equivalent fractions though the fraction appears to the same.

3/5 = 6/10 = 9/15 = 12/20

x2 x3 x4

- When adding and subtraction fractions with different denominators learners should be made aware that they will use the LCM e.g. in 1/3 + 1/4 the LCM is 12 such that:

  a) 1/3 +1/4
  = $\frac{4+3}{12}$
  = 7/12

b) 1/3-1/4

$= \frac{(4x1)-(3x1)}{12} = \frac{4-3}{12}$

$= 1/12$

NB: When adding or subtracting fractions, only numerators are added.

❖ Writing percentages as fractions and vice-versa

20 % as a fraction is 20/100 = 1/5

3/4 as a percentage is 3/4 x 25/25 = 75/ 100 = 75 %

TOPIC: Decimals

Specific Objectives 1.4.1.1 to 1.4.1.8

Content

Concepts: tenths, hundredths, decimals numbers, decimal places

Support Materials

Abacus, dienies, unifix from 10s to 1000s and fraction charts.

Suggested Teaching / Learning Activities

❖ Let learners denote place value from tenths to hundredths.

H	T	U	t	h	
	4	6	8		denotes 46.8
		7	2	1	denotes 7.21
3	2	9	4	7	denotes 329. 47

❖ Show conversion of fractions to decimals by **long division**

8/10 = 8 ÷10 = 0.8

36/100 = 36 ÷ 100 = 0.36,

Hence the relationship between a decimal and a fraction

❖ Let learners carryout the four basic operations on decimals up to the tenth place value.

Example:

0.2 + 34.1 = --------

34.1 – 0.2 = -------

6.1 x 2 = --------- 18.3 x 10 =-------

0.7 x 6 = ---------- 0.6 x 100 = -------

6.8 ÷ 2 = ---------
18.9 ÷ 9 = ------

Betty's height is 1.3 m and Kago is 1.5 m tall.
What is their height together?
How tall is Kago more than Betty is?

NB: PLEASE NOTE THAT DECIMAL POINTS MUST BE USED NOT DECIMAL COMMAS**.**

TOPIC: Money

Specific Objectives: 1.5.1.1 – 1.5.1.9

Content:
Concepts: Money, conversions of money, purchasing, bills

Support Materials:
Money: coins and notes, invoices and receipts.

Suggested Teaching/Learning Activities:

- ❖ Let learners interpret national symbols on coins and notes and discuss their significance:

Example: Coat of Arms
*The **three cogwheels** stand for industry*
*The **bull's head** symbolises the cattle industry*
***The three wavy blue band**s represent reliance on water*
*The **black and white stripe**s of zebra stand for the same racial co-operation*
*The **two zebras** and the **elephant tusk** represent the county's natural fauna*
*The **head of sorghum** stands for agriculture*
*The motto **Pula** means "let there be rain'.*

- ❖ Let pupils discuss shopping experiences e.g. Christmas shopping

- ❖ Make a shopping corner and let learners play shop, where addition, subtraction, multiplication and division of money are involved.

- ❖ Show real copies of invoices, bills and receipts for learners to study, discuss and make their own.

Example of an invoice:

Molefhe's Supermarket Bought by:---------- Date: / /			
Quantity	Items	P	t
1	350 ml milk carton @ P52.30 each	52	30
2	2 l cooking oil @ P14.25 each	28	50
5	Packets of sweet-aid @ P0.75 each	3	75
	Total	84	55

PROBLEM AREA

The difference between an invoice and a receipt:
Invoice is detailed and provides full information and specific service provided
Receipt is a proof of payment - it verifies payment.

INFUSION

The topic money reinforces entrepreneurial skills. Teachers could emphasise the importance of starting one's business to earn a living.
The discussion on money symbols realises vision 2016 pillar number 7 (*A united and proud nation).*

Module 2: Geometry

Module Overview:

This module reinforces what learners learnt at lower primary. The learners will extend their knowledge relating it to real life situations that develop further understanding of parallel, perpendicular and curved lines.

Learners will gain knowledge on angles by naming, measuring and drawing them. They will identify lines of symmetry through a variety of activities. They will explore their local environment to identify tessellating objects. The learners will also learn basic concept on transformational geometry and co-ordinates.

Topic: Lines

Specific objectives: 2.1.1.1. – 2.1.1.5.

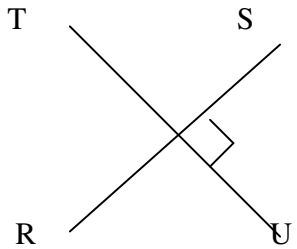
Content:
Concepts: Perpendicular, parallel and curve lines, compass, ruler, set square, Protractor
Skills: Drawing

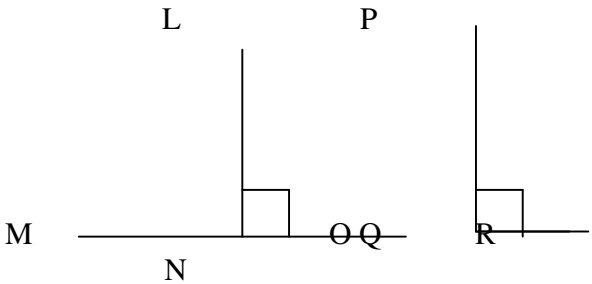
Support Materials:
Match sticks pairs of compass, ruler, string, cones, and balls.

Suggested Learning/Teaching Activities:

- ❖ Identify perpendicular lines in the classroom such as on the window frames, ceiling, and chalkboard. *(Environment is infused here).*

A. **Perpendicular** – *Lines that meet at a right angle.*





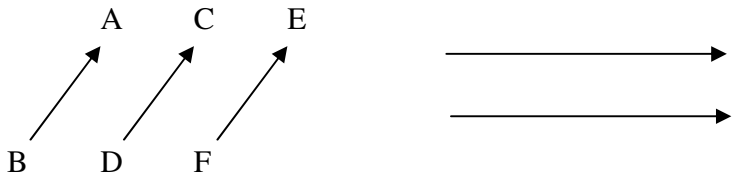
LN is perpendicular to MO PQ is Perpendicular to QR

- Highlight the symbol of a right angle. 

- ❖ Identify parallel lines in the environment such as on the floor tiles, windows, and chalkboard.

B. **Parallel Lines** – *lines that do not meet.* The distance between the lines should be constant.

Example

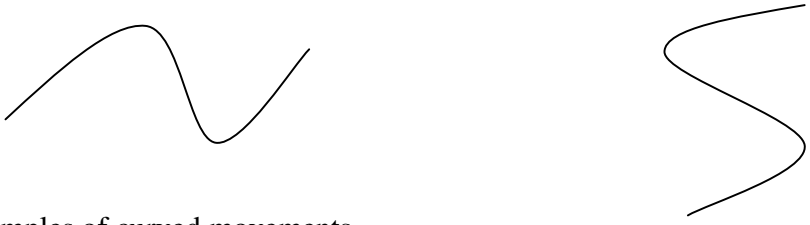


Note: Make learners aware that they can check if lines are parallel by visual inspection and/or by measuring.

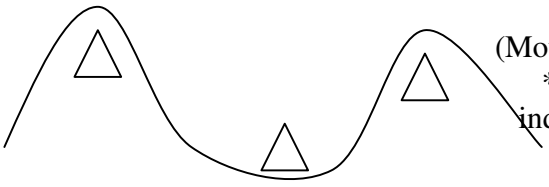
Show an example of how to measure parallel lines.

- ❖ Identify curved lines in objects within the environment e.g. of drawings, paintings, furniture, the rainbow, moon

C. Curved Lines:

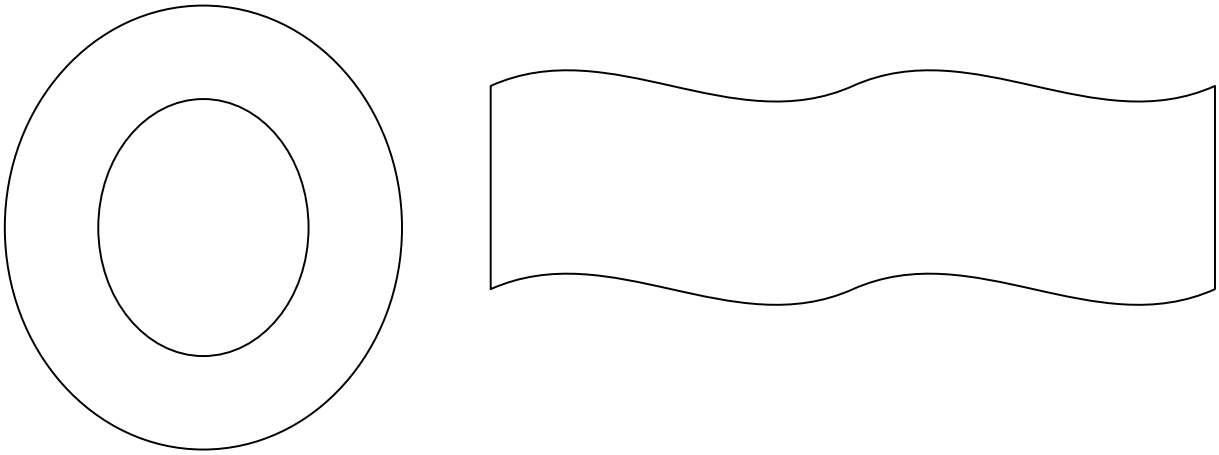


Give examples of curved movements.



(Moving around cones)
* Improve the drawing to clearly indicate this.

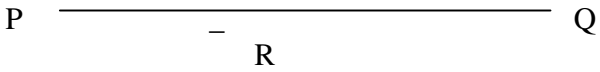
Include curved lines that keep the same distance. E.g. rail road.



❖ Drawing Perpendicular Lines

Step1: Draw a line and name it PQ.

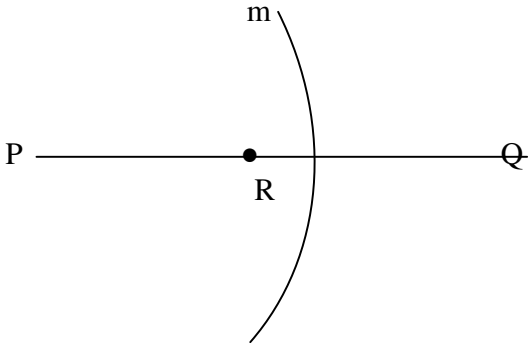
Mark the mid-point R.



Drawing perpendicular lines using pairs of compass.

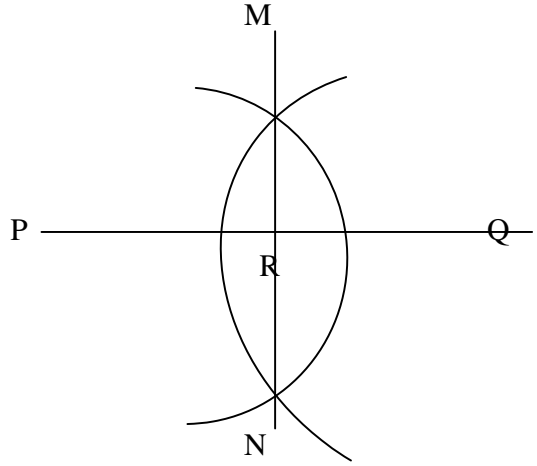
Place the activity in page 13 here (using a compass to make patterns – correct illustration.

Step 2: Place the compass point at P and the pencil beyond the mid-point more than half the line. Draw an arc through PQ.



Step 3: Draw another arc through P to meet the Q arc at the top and bottom of line P. Name the points of intersection m and n.

Now place the compass at Q maintaining the same opening. Draw another arc though line PQ. Name the points of intersection M and N.



Step 4: Draw a line to join points M and n as shown above. Therefore line MN is perpendicular to PQ

Let the learners use the set - square to verify. The students should be advised to use safety precautions when handling the compass.
Note that when using a compass:

The point of a pencil should be in line with the point of a compass
The pencil should not be too long
Hold the compass by the tip.

❖ Perpendicular lines may also be drawn using set - squares and protractors.

Topic: Angles

Specific Objectives: 2.2.1.1 – 2.2.1.6.

Content:

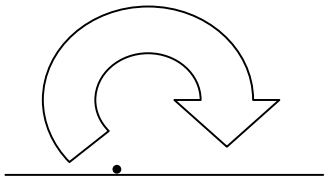
Concepts: Angles: straight, reflex, right, revolution, clockwise and anticlockwise turns
Skills: Measuring, drawing.

Support Materials:
Protractor, pairs of compass, globe, dice, manila sheet, card board.

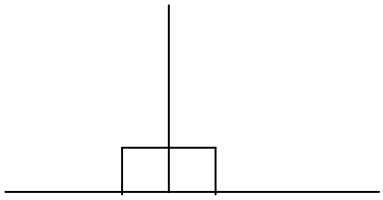
Suggested Learning/Teaching Activities:

A) Straight angle

Have a point on the straight line where the angle will be measured.



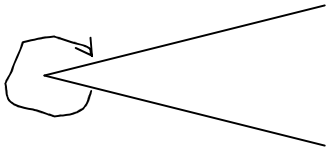
A straight angle is a 180º angle or 2 right angles side by side.

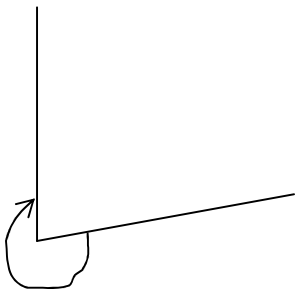


- Identify straight angles on objects within the environment.

B). Reflex angles

A reflex angle is an angle that is greater than a straight angle (180º) but less than revolution (360º).

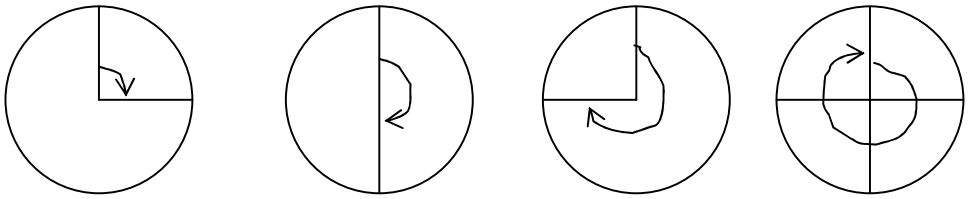




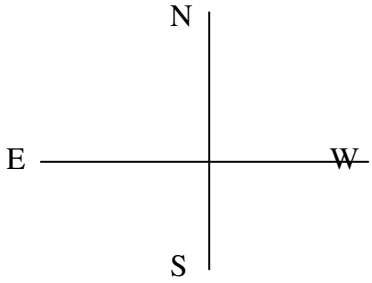
- Use the arms of clock face to show the angles.
- Identify angles formed by the brunches of stems.

C). Revolutions and fractions of a revolution.

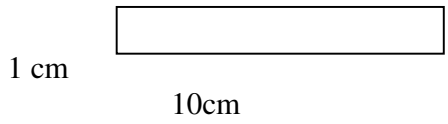
- Use a clock face by moving the hands clockwise, to show 1/4 of a revolution = 90º, 1/2 of 360 = 180º, 3/4 of 360 = 270º and a complete revolution.



- Clockwise and anticlockwise turns in cardinal points. Familiarise learners with cardinal points, East, South, North, West.



Use cardboard strips to show the clockwise and anticlockwise turns – cut two 1 cm x 10 cm strips.



Punch a hole near one end of each strip and fasten them together. Keep one strip tight and rotate it to the desired turn or cardinal point – clockwise or anticlockwise.

- Learners may play a Cardinal Game to reinforce their understanding of the relationship between cardinal points and angles.

  A game played by 2 teams of maybe 4 pupils/learners.

  One learner from Team 1 asks a question like 'East clockwise West how much angle have I covered? Team 2 must tell the angle formed.

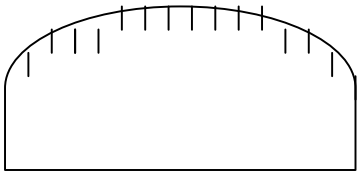
  Team 2 will also ask a question, e. g. West 270º anticlockwise where do I end? Team 1 then will answer.

NB: The team will continue asking each other for the given time. Scores must be kept for a correct answer. At the end of the set time, the scores are added and the team that has scored more wins.

❖ Measuring using a Protractor

A protractor has two scales, the outer and the inner scale. Use only one scale in one measurement i.e. if you measure using the inner scale you should use it to the end.

The baseline is not used for measurements. Use the 0º- 180º line.



Place the centre of the protractor line 0º- 180º on the vertex of the given angle. The 0º line should be in line with one arm of the angle.
Read from 0º to the other arm of the angle. The arms of the angle may be extended.

Topic: Quadrilaterals

Specific Objectives: 2.3.1.1. – 2.3.1.7

Content:

Concepts: quadrilaterals, equal opposite angles and sides, different opposite angles and sides, properties
Skills: Sorting, drawing, investigating, recording.

Support Materials:

Cut-out shapes, square grid, pairs of scissors, rulers, pencils

Suggested Learning/Teaching Activities:

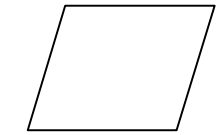
❖ Discuss properties of a quadrilateral

- Plain flat shape
- Has four sides
- Interior angles add up to 360º.
- Can be divided into 2 triangles.

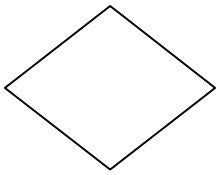
There is need for the meaning of the word quad to be emphasised. Let learners understand that it means **four.**

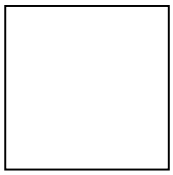
❖ Provide learners with many different shapes such as square, kite, triangle, and circle and let them sort the quadrilaterals.

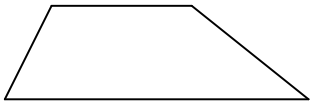
Examples of Quadrilaterals

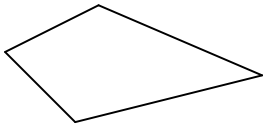

Parallelogram

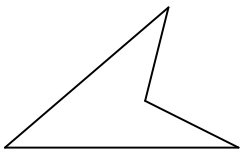

Rectangle


Rhombus

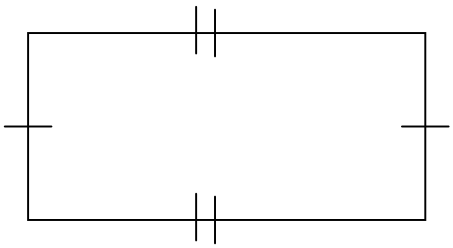

Square


Trapezium


Kite


Composite quadrilateral

❖ Equal opposite angles and sides. Provide pre-cut quadrilaterals to learners. Mark the opposite sides and angles differently e.g.



Fold, measure or trace to compare the sides and angles.
Each learner should have a table to record.

Shape	***Equal Opposite Angles***	***Equal Opposite Sides***
Square Kite Rhombus	√	√

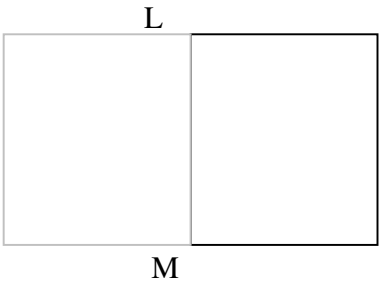
Topic: Symmetry

Specific Objectives: 2.4.1.1 – 2.4.1.6.

Content:
Concepts: Lines of symmetry, two dimensional shapes, and reflection
Skills: drawing, sorting, and matching

Support Materials:
Manila, pairs of scissors, paint, plain papers, leaves, chair, table, mirrors.

Suggested Learning/Teaching Activities:
A line of Symmetry is a line that divides a shape into 2 equal parts that do not overlap.



Line LM divides the rectangle into 2 equal parts.

❖ Let learners cut out different shapes and fold to find their lines of symmetry.

Paper folding/ blotting, painting/ Ink blotting.

❖ Write the letters of the alphabet in capital letters. Cut them and fold to check if they do not overlap

A B C

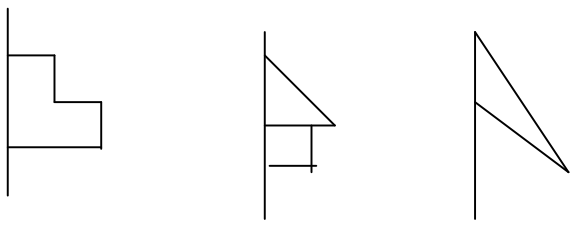
❖ Let learners find lines of symmetry in different shapes and objects in the environment, e. g. tree leave.

❖ Picture Matching.

- Draw pictures and cut them symmetrically.
- Mix the pictures and let learners match.

❖ Completing Figures.

Provide half-drawn pictures and let the learners complete them:



If learners find it difficult, allow them to use the mirror.

Note: The parts of a shape cut symmetrically should not overlap.

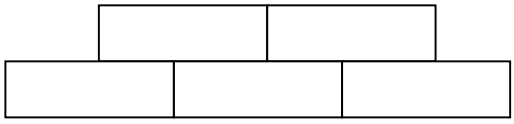
- ❖ Take a nature tour to observe symmetry in insects, butterflies, leaves flowers animal footprints etc.

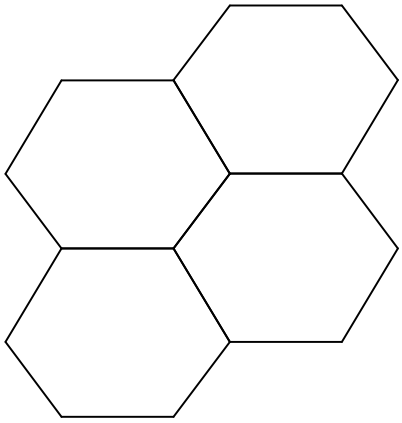
Topic: Tessellation
Objectives: 2.5.1.1 – 2.5.1.2

Content
Concepts: tessellation

Suggested Learning/Teaching Activities:

- ❖ There are a lot of tessellation in the environment such as tiles, buildings, clothes and patterns in our traditional artefacts.
- ❖ Make tessellation of different shapes





Note: that a tessellating object is

Topic: Transformation

Objectives:

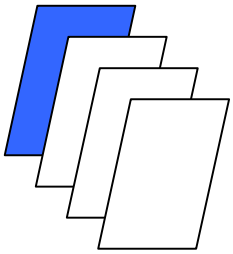
Content
Concepts:

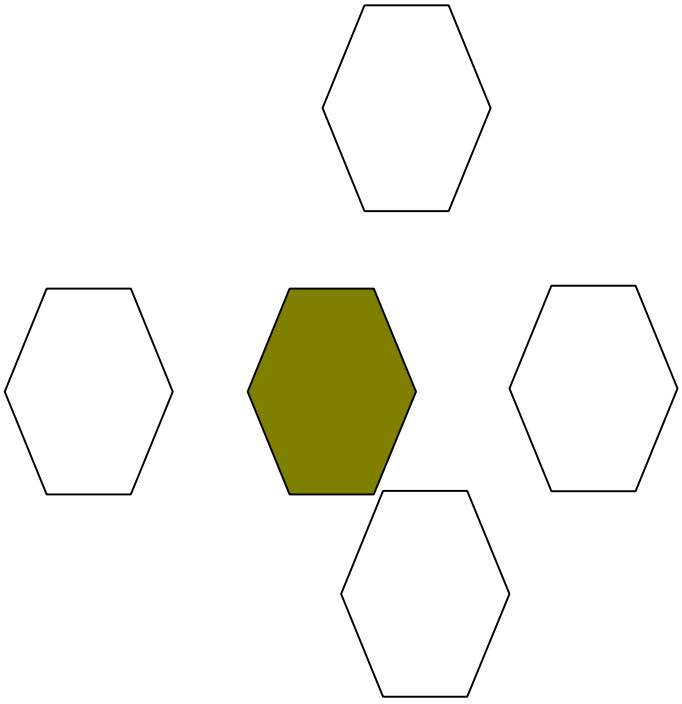
Suggested Learning/Teaching Activities:

- ❖ Introduction of translation, reflection and rotation would be in the form of the movements the transformation represents such as slides flips and turns respectively. Teachers can demonstrate these using real objects.

❖ Slides represents movement of an object either horizontally to the right or left, vertically up or down or a combination of a horizontal movement, right or left followed by a vertical movement up or down.

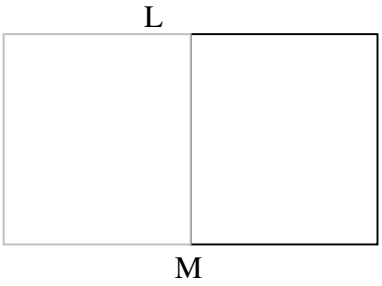
Example:





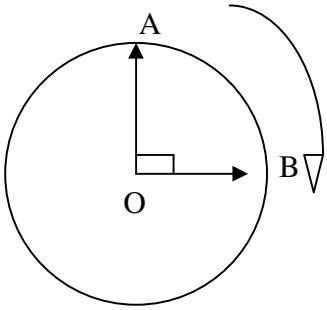
❖ Flips are movements whereby the object is flipped to the other side such that it is exactly the same on the other side of the line, normally referred to as mirror line.

Example:
The line LM is the mirror line after flipping the middle page of an exercise book.



Let learners observe a butterfly and discuss flips and the symmetry that it displays.

Turns
Turns can be explained easily with the demonstration of how we build our traditional hut or houses. When we start, a nail is tied to a rope and pinned to a certain point, which is the centre. Then somebody moves the nail around ending up with a circle.



The figure shows a **right** turn from point A to point about the centre O.

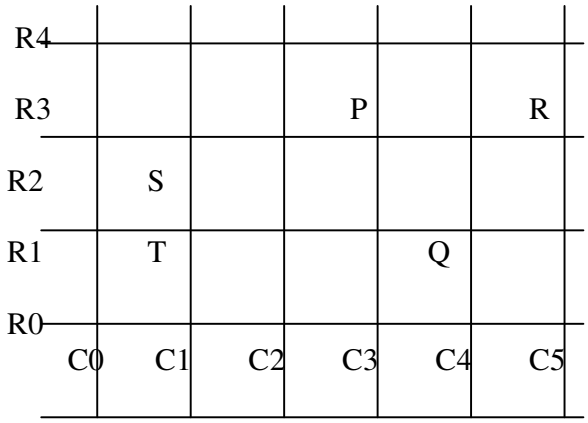
Topic: Coordinates

Specific Objectives: 2.7.1.1 – 2.7.1.4

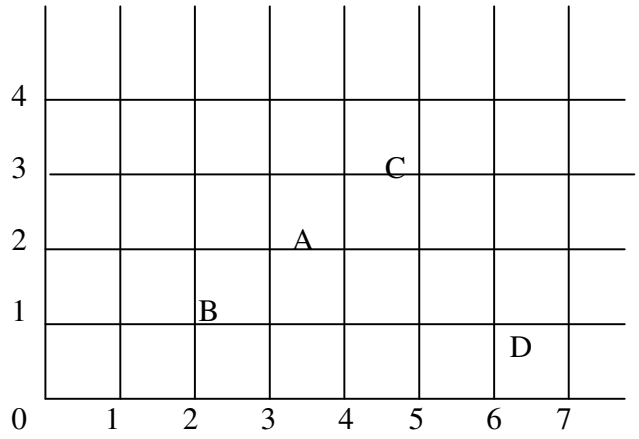
Content
Concepts:

Suggested Learning/Teaching Activities:

❖ R is in position C5R3. State the position of P, Q, S, T



To get to position A in the diagram below, one moves 4 steps to the right and 2 steps up.



Describe how one can get to position B, C, and D.

MODULE 3: Measures

Module Overview:

The Module intends to expose learners to the handling and using of appropriate measuring instruments as well as reading and interpreting numbers and scales with some level of accuracy. Learners will choose appropriate standard units of length, area, capacity, volume, mass and time and make sensible estimations of them in real life situations.

Measurement is a practical activity and learners need to develop the measuring skills, which they should extend to other subjects in the curriculum. They will also apply measurement formulas, carryout conversions and solve word problems.

Topic: Length

Specific Objectives: 3.1.1.1 – 3.1.1.6

Content:
Concept: Instruments and units of length, perimeter, diagonals, boundaries, Conversions, Millimetre, Cent-meter, Meter, Kilometre, linear dimensions, curves, circles, irregular shapes
Skill: Measuring, estimating, calculating.

Support Materials:
Measuring instruments: Ruler, Tape Measure, String, and Click Wheel.

Suggested Learning/Teaching Activities:
- Let learners measure perimeters of circles and irregular shapes using a string and a ruler

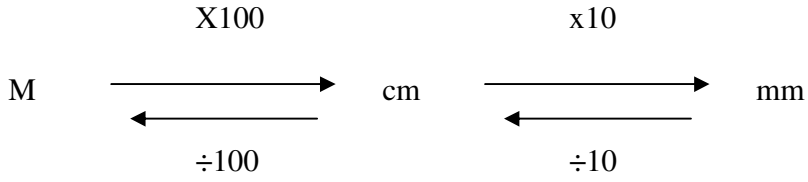
- List objects of linear measurement, appropriate instruments and unit of measure.

Example:

Object		**Unit**	**Boundary**	**Diagonal**
Pencil	ruler	**cm/mm**		
Table	Meter stick	**M/cm**		
Book	Ruler	**cm/mm**		
Football pitch	Meter stick	M		

- Let learners use the formula (L + W + L + W) or (S + S + S + S) to find perimeters of rectangles and squares.

- Let learners convert **mm** to **cm** to **M** vice versa.



Topic: Area

Specific Objectives: 3.2.1.1. – 3.2.1.3

Content:
Concepts: Area, triangles, and irregular polygons
Skills: Estimating, calculating.

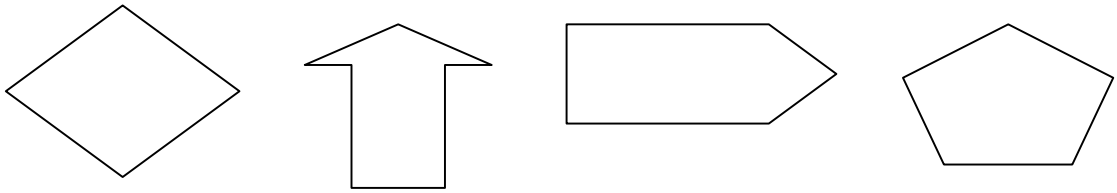
Support Materials:
Square grid, pencils, and polygons.

Suggested Learning/Teaching Activities:
- Let learners calculate the areas of squares and rectangles using formula.

- Let learners estimate areas of irregular polygons.

Example:

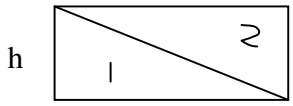
Draw and cut out irregular polygons:



Trace the polygons on a square grid. Count the squares to find appropriate area.

❖ (Using the rectangles to lead to the formula (1/2b x h)





This could be done practically (by manipulation) to show that the area of triangle is 1/2 the area of rectangle, hence A = 1/2(b x h).

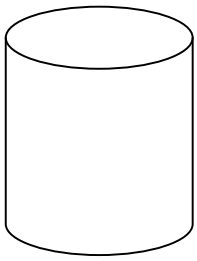
Topic: Mass

Specific Objectives: 3.3.1.1. – 3.3.1.4

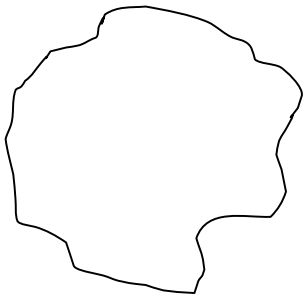
Content:
Concepts: Mass, kilograms, grams
Skill: Matching, Sorting.
Support Materials:
Rialia-Stones, buckets, Scales, Weights

Suggested Learning/Teaching Activities:

❖ Bring different object such as stones, buckets and dusters. Provide a box of word cards showing kg and g. Pick the word cards and match them in the objects provided.



Kg



g

Measure the mass to confirm the units.
Conversion table.

Object	Kg	g
Stone	2	2000
Sugar		1500
Packet of tea	0.6	

❖ Carry out practical activities on mass and carryout basic operations on mass.

Problem Area:

Differentiate between mass and weight (see glossary)

Topic: Capacity

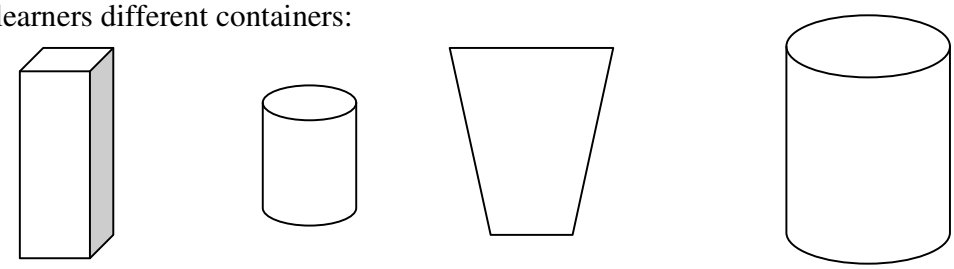
Specific Objectives: 3.4.1.1. – 3.4.1.5

Content:
Concepts: Capacity

Support Materials:
Containers, Litter Bottles, Measuring Cylinder

Suggested Learning/Teaching Activities:

❖ Give learners different containers:



2l **1l** **5l** **10l**

Provide word cards showing appropriate capacities. Let learners match the cards with the containers. Let learners measure the capacity to the nearest litre.

❖ Use all four operations to solve work problems involving capacity.

Problem Areas:
Capacity and volume should not be confused.

Infusion:
Teachers could highlight the effects of littering and the advantages of recycling reuse and reduce and give examples of the containers given or used by the learners.

Topic: Volume

Specific Objectives: 3.5.1.1 – 3.5.1.4.

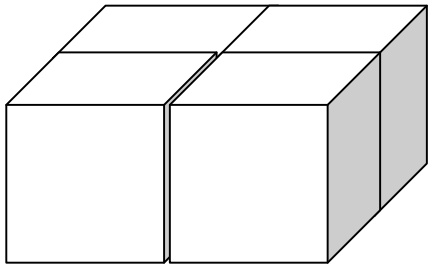
Content:
Concepts: Volume, cubes and cuboids.
Skills: modelling.

Support Materials:
Boxes, Cubes

Suggested Learning/Teaching Activities:

- Let learners make models of cubes and cuboids of given volumes.

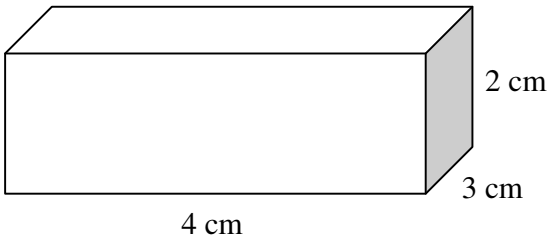
- Learners fill given containers with cubes and count them.



4 cubes

Formula for finding volume.

One needs to know the length, width and height of a shape.



L = 4cm
W = 3cm
H = 2cm

Multiply the given dimensions i.e. L x W x H, e.g.

V = L x W x H
V = 4cm x 3cm x 2cm
V = 12cm x 2
V = 24cm3

Topic: Time

Specific Objectives: 3.6.1.1. – 3.6.1.8

Content:
Concepts: Minutes and hours, Time span.
Skills: Estimating, Calculating, Measuring.

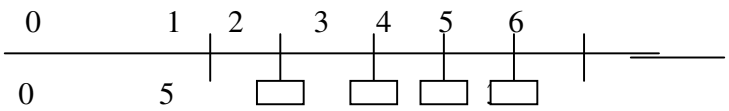
Support Materials:
Clock face – analogue.

Suggested Learning/Teaching Activities:

- Let the learners read the minutes from a clock face.

- Let learners convert hours to minutes and minutes to hours.

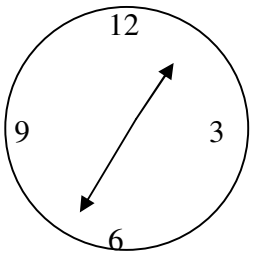
Example:
Time line

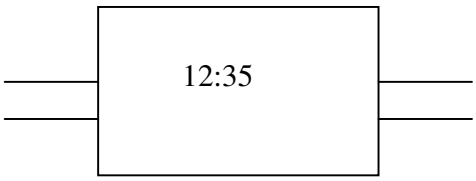


Note that learners move the minute hand to the numbers 1,2, up to 12 and count the number of minutes from 0 to 60.

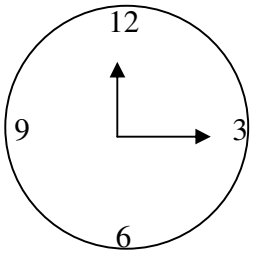
- Demonstrate how to read an analogue clock to the nearest minute.

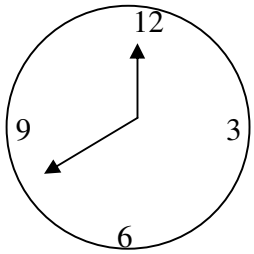
Read the times shown below.





- Estimate time intervals given the initial position of the minute hand and the last position. Use the clock face to move the minute hand to confirm the above estimate.





NB: Round down 1 sec to 29 sec
Round up 30 sec to 59 sec

- Identify time span:

Year	Decade	Century
12 months	10 years	100 years

Evaluation:

- Learners discuss times for TV and Radio programmes and their duration, even the presenters for those programmes. This would be a way of infusing the world of work in Mathematics.

Module 4: Problem Solving

Module Overview

The main purpose of this module is to help learners to always try different ways and alternative approaches of overcoming difficulties when solving problems of different kinds. The problems could be derived from numbers, geometry, measures, practical activities or non-routine real life situations. The module also provides a basis for the effectiveness use of mathematics as a tool in a wide range of activities within school and outside school life. The module includes games and puzzles as well as investigations, which are the means by which learners can derive pleasure and enjoyment and develop interest and enthusiasm on mathematics, at the same time developing skills such as problem solving, investigative, communication and co-operative. The module also provides learners with opportunities to use their own expertise to find their own ways through problem solving and investigations where strategies, methods and solutions are not immediately obvious. This is where learners need to show initiative, creativity and flexibility in their approach to problems.

Topic: Games and Puzzles

Specific Objectives: 4.1.1.1 - 4.1.1.2

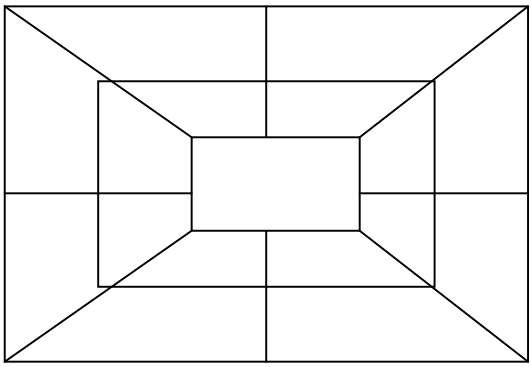
Content
Concepts*: Games, puzzles, and non- routine problems*
Skills: *Problem solving*

Support Materials
Stones, dice, bottle tops

Suggested Teaching /Learning Materials

- Play mathematical games such as Dominoes, bingo, monopoly, *morabaraba*, *mhele*
- Solve mathematical puzzles like jigsaw, magic square, number and word searchers, mazes
- Carry out non routine activities involving numbers, geometry and measures

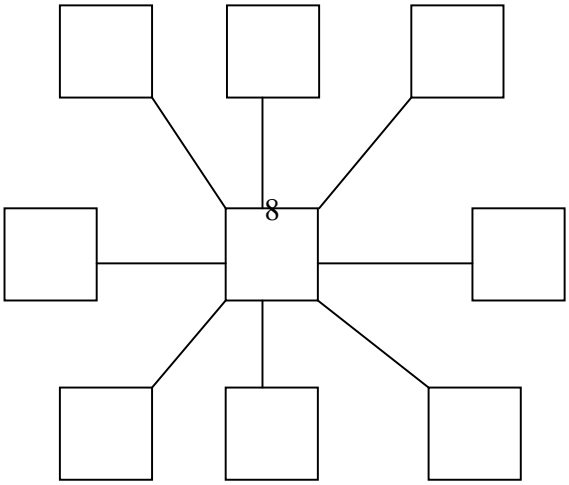
Examples of:
i). Games
Two players using different objects like stones and beads play the game in turns. A point is made when one player put three of his/her objects in a line on a figure like the one below. The figure can be drawn on a hard board or on the ground. When one makes a point he/she gets the other players object. The winner is the one who is able to accumulate more of the opponent's objects.



ii) Puzzles

a) Number chain

Use numbers from 1 to 9. Fill the squares with numbers such that each chain of three numbers adds up to 19.



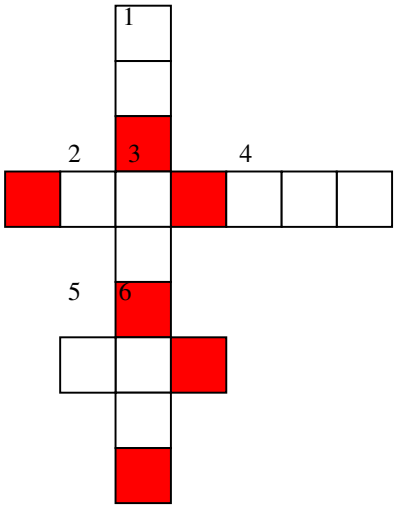
b) Cross Number puzzle

Complete this number puzzle

Clues across
2. The sum of 9 and 85
4. Two more than ninety-eight
5. The product of 24 and 3

Clues down
1. Seven less than a hundred
3. Divide 65 by 5

6. 100 subtract 71



c) Word Search

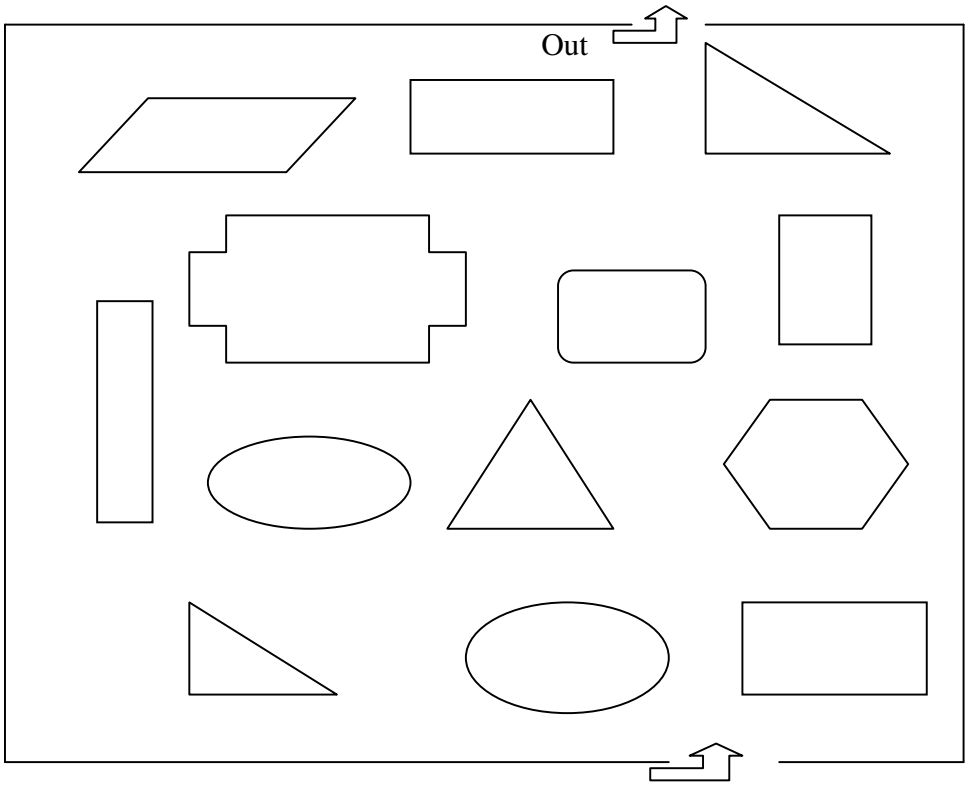
Search for these words and circle them. *Area, Perimeter, Circle, Cuboid, Cube, Triangle, Rectangle and Square*

The words can be from left to right, top to bottom, diagonally from bottom to top and diagonally from top to bottom.

G F J P E R I M E T E R A S E B O E R D E A B E A
P C G E F X D T W T E B P U X R I P K N L C D B
X I U B E U L U G C Y U H X A I I R K E W F G C
D R I B H X S E W Y K A U C U B E L C X H I D U
F C H I O D R I N K S A Q U V I Q C O I L T L O E
L L K N I I E U L O E S I I R E M O T X K H R S I F
B E S T F C D T R I A N G L E D G A I L E M K E G
I M E K I B J I A R E P L S L A S T N R G E I D C H
NO L R D U R L T B C U Y E O H I G N Y T R O I L
DS U F T E U R E W T L A L E L G L Y O N K P I W
Z H Z E A P U O E J N Y X C I L N E P E G E S A K

d) Maze

Enter the maze by tossing a coin. Heads go right and tails go left. Toss the coin at each junction to decide on which way to go and trace your path. Repeat to find out if you would go through the same path.



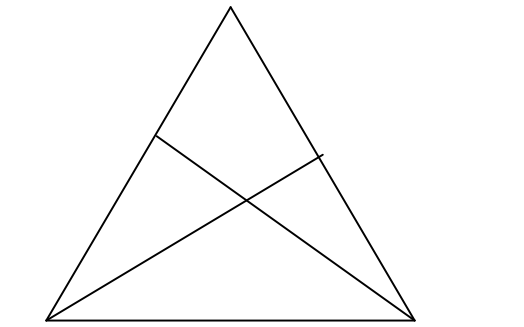
e) Magic Square

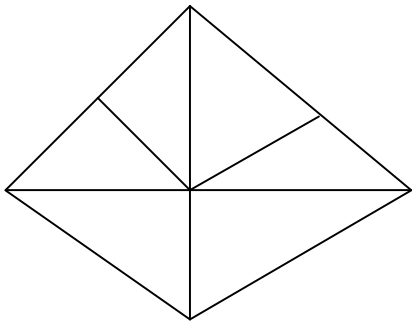
The magic square adds up to 97 in each direction. Copy and complete it.

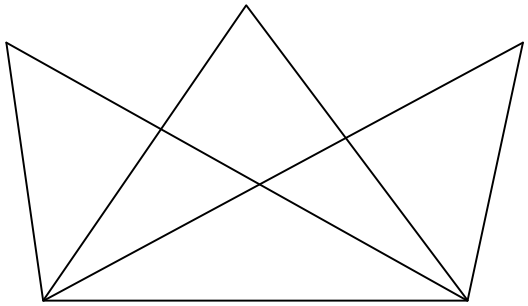
	29	
	16	
2		

i) Non- routine activities and Challenging questions

a) How many triangles are there altogether in each drawing?







b) There are 87 children and 7 teachers to accompany them to a school trip. They are using buses that can carry only 25 people. How many children and teachers will each bus carry?

c) A container holds 67 litres of oil. A bottle holds 5 litres of oil. The container is full of oil and the bottle is empty.

How many empty bottles can be filled from the container?

How much oil is left in the container?

d) Masego counted her pigs and chickens. Altogether they were 11 and they had 30 legs. How many of each of the animals, that is pigs and chickens were there?

e) The numbers: 12, 6, 17, 23 and 7 are scores five BDF football players have ever scored.

What is the total score of the players?

Find three different ways of making a score of 29 with two or three of the numbers

Is it possible to score 56 with the numbers?

Topic: Investigations

There are no clear distinctions between problem solving and investigative work. However, in investigative approach to teaching and learning, learners are more at liberty to think of alternative strategies in a broader way in order to come up with a solution, hence problem solving can require an investigative approach and an investigation can end up being a project. On the other hand, activities for both problem solving and investigative work can have a range of possible outcomes, no solution at all, a unique solution and a solution provided a lot of information is made available. Teachers are therefore encouraged to provide learners with a variety of such problems or activities. They are also to encourage learners to express their interests and ask questions as by so doing it motivates them.

Specific Objectives:
4.2.1.1 carry out simple investigations involving numbers, geometry and measures
4.2.1.2 make mobiles from shapes

Content
Concepts: *investigating, discovering*
Skills: *investigating, pattern making*

Support Materials

Suggested Teaching/Learning Activities

Investigations

There are different kinds of investigations, examples of which are listed below. There are many others that teachers are encouraged to come up with.

a) Palindromes
A palindrome is a number that reads the same in both directions, for example, 44, 232, 35653. The method for producing palindromes is given below:
1. Write down any number with two or three digits
2. Write down the digit of the number in reverse order
3. Add up the two numbers written down
4. Write the answer
5. If the number is a palindrome you stop.
6. If the number is not a palindrome you again write down the answer you got in reverse order and continue from step 3.

Example:
1. The number chosen as input is 271
2. The reverse is 172
3. Add 271 and 172
4. The answer is 443

5. The answer is not a palindrome
6. The number in reverse order is 344
7. Add 443 and 344
8. The answer is 787
9. The number is a palindrome.

Find three input numbers that will give the same palindrome
Find two digit numbers that will take the longest to produce a palindrome

b) Number patterns
There are a number of patterns or investigations that can be made with the number 9. Teachers are encouraged to search for various sources for reference.
Example:

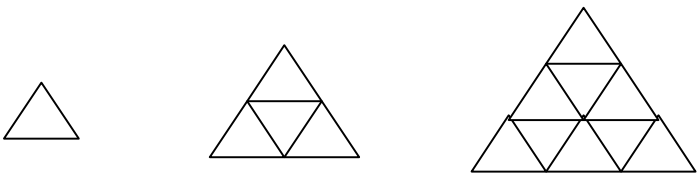
i) 1 x 9 + 2 =11
12 x 9 + 3 = 111
123 x 9 + 4 = 1111

ii) 6 x 2 = 6 x 1 x 2
6 x 4 = 6 x 2 x 2
6 x 6 = 6 x 3 x 2

Continue with the patterns to the tenth step.
What can you say about each of the patterns?
Learners should be encouraged to explore other similar patterns.

c) Permutations
Kabelo has to choose a skirt and blouse of three different colours, black, white and red. How many combinations can he make?
Green skirts and blouses are introduced. How many combinations can he make with four colours?
Suppose the colours are increased to six, with blue and yellow and this time Kabelo has to make combinations for a skirt, blouse and a pair of shoes. Investigate to find how many he can make.

d) Geometric Patterns

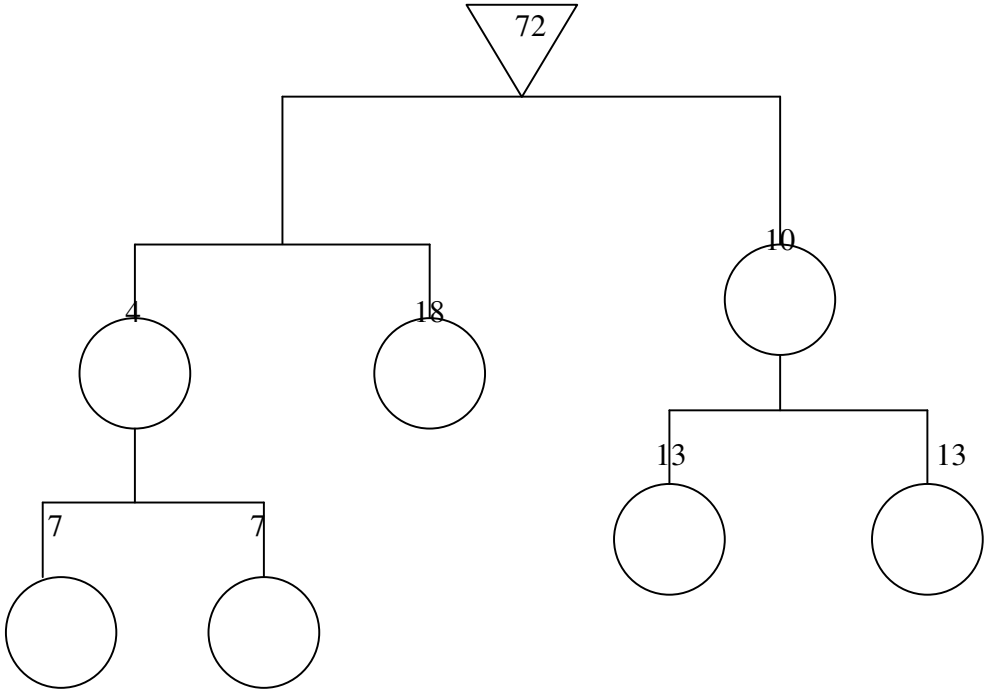


9 sticks are used in the second pattern
How many sticks are needed for the fourth pattern?

How many triangles are there in the third pattern? (Note that there are at least 13 triangles including the big ones)

Mobiles

Number mobiles



In the diagram, the left hand must balance the right hand side.
Make your own diagrams and investigate a range of mobiles with the same total as the one above
Let learners choose their own mobile totals
Give learners more challenging mobiles

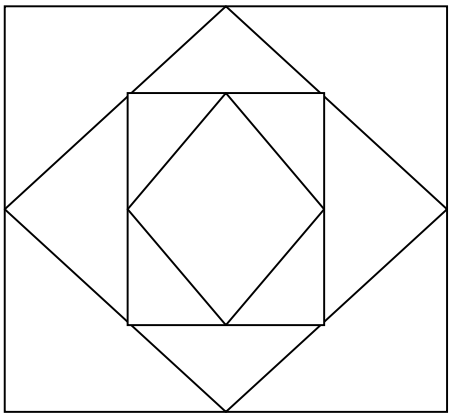
Evaluation

Draw a large square using a grid / squared exercise book.
Put a point at the mid point of each side and join the dots to make another square

Continue the process.

i) Find the measurement of each square formed
ii) Predict the measurement of the eighth square

iii) Formulate patterns for different ways of colouring the squares as a group



Problem Area

It must be noted that Problem Solving activities can be infused in other topics where possible, otherwise the Problem solving topics should be taught just like other topics in the syllabus. The module is not optional.

- It is important to use mathematical terms appropriately. Teachers are encouraged to observe learners especially in discussions and presentations. For example, using sums for addition, product for multiplication and reading numbers after the decimal point as fractions not as whole numbers, like 23.11 is to be read as twenty three point one, one not twenty three point eleven.

MODULE 5: Statistics

Module Overview

The module intends to instil in learners the different ways of disseminating information. The learner's ability to read and interpret information in tables, graphs and charts is very important in understanding the world around them. Learners will also collect and record information in tabular form, which they later represents in pictographs, bar charts and line graphs. They learn about the Measure of Central Tendency such as mode median and the range of a distribution.

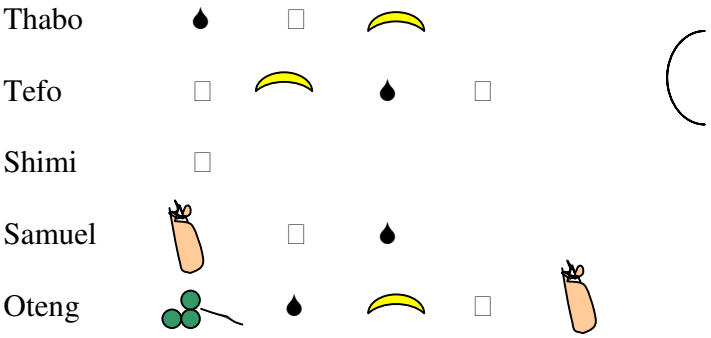
Topic: Graphs
Specific Objectives: 5.1.1.1 – 5.1.1.4
Content:
Concepts: Data, bar graphs, line graphs

Support Materials:
Charts, pictures, objects

Suggested Teaching/Learning Activities:

- Display data in a pictograph

Example:
Pictograph showing the number of fruits eaten by five people in a week:



Which fruit was mostly eaten that week?
How many different types of fruits were eaten?

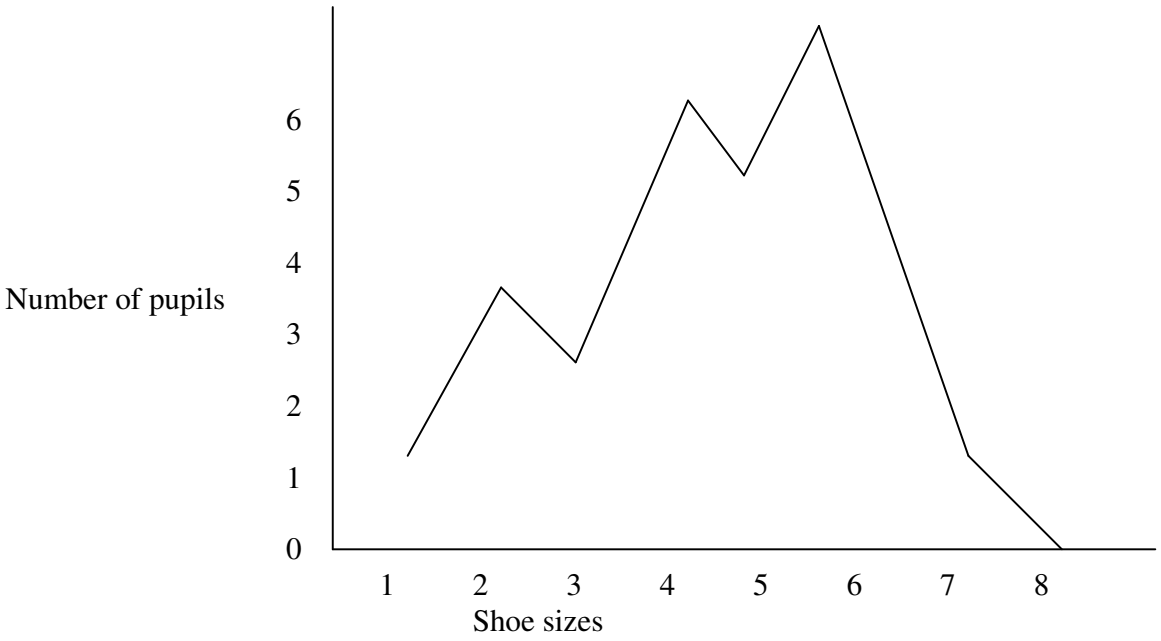
- Allow learners to collect data and record it in tables using tally marks e.g. colour of cars, age of pupils, shoe sizes/co lour etc

Examples:

Sizes of shoes worn by pupils:

SHOE SIZE	TALLY MARKS	NUMBER OF PUPILS
1	1	1
2	111	3
3	11	2
4	~~1111~~	5
5	1111	4
6	~~1111~~ 1	6
7	1	1
8		0
TOTAL		22

Line graph showing the shoe sizes worn by pupils



Which size did most pupils wear?
How many pupils wear size 4 and above?

Topic: Measures of Central Tendency:

Specific Objectives: 5.2.1.1. -5.2.1.4

Content:
Concepts: mode, median, range, distribution.

Suggested Teaching/Learning Activities:

- Provide learners with an odd numbered distribution so that there is only one median. Example:
  6, 1, 3, 4, 3, 2, 3, 6, 5

In the distribution given, what is the mode, median and the range?

Mode = 3

To find the median first arrange the distribution in order, 1, 2, 3, 3, 3, 4, 5, 6, 6
Median = 3 (the number in the middle)

Range = 6 – 1 = 5 (the difference between the smallest number and the biggest number in a distribution.)

Topic: Data Collection and Analysis

Specific Objectives: 5.3.1.1. -5.3.1.5

Content:
Concepts: Project, data, display, interpret

Support Material:

Suggested Teaching/Learning Activities:

- Identify major problems in the community concerning HIV/AIDS and Environmental Education issues, for example, the number of patients who take ARV treatment by age. Let learners collect data and tabulate it. They should draw a suitable graph to represent the data. They should also suggest possible solutions to the problem.

Note that objective 5.1.1.1 about HIV and AIDS and Environmental Education is compulsory. This is a way of infusing or integrating emerging issues in Mathematics.

Topic: Probability

Specific Objectives: 5.4.1.1. -5.4.1.2

Content:
Concepts: Events, never, sometimes, always, certain, possible impossible

Support Materials:
Dice, Coins, playing cards, coloured balls

Suggested Teaching/Learning Activities:

- Let learners play games whereby one person wins, e.g. spin, cards, dice, *mhele wa seraro* (three). Discuss the results e.g. who won, who lost, how often did one win?

- Let learners discuss and list events that involve people winning, loosing and events that involve predictions e.g. weather, elections, sport etc.

- Classify/group the listed events as never, sometimes, always, certainly, possible, impossible.

Events	Never	Sometimes	Always	Unlikely	Certainly	Possible	Impossible
The sun rises from the east			**X**				
Tomorrow is my birthday					**X** (if ones birthday is the following day)		**X**
It is going to rain (No sign of clouds)				**X**			
Chickens fly		**X**					
The sun will rise from west tomorrow							**X**

NB: These words are used to tell the changes of success or failure and loss or win.

Module 6: Algebra

Topic: Algebra

Specific Objectives: 6.1.1.1. – 6.1.1.9

Content:
Concepts: patterns, linear sequence, consecutive terms, linear geometric, arrow Diagrams, number sentences, expressions

Support Materials:

Suggested Teaching/Learning Activities:

- The 100 table can help generate as many patterns as possible.

Example 1:
Consider the faint numbers from the table: 34, 45, 56, 67.
What are the next two numbers in the sequence?
What is the rule for this pattern? (Add 11 to each number)

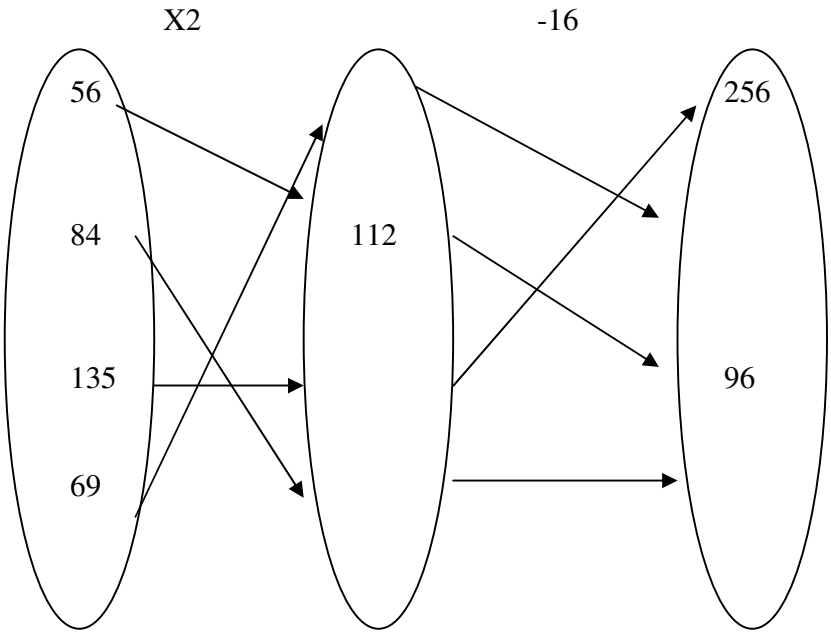
1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	57	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Example 2:
Consider the pattern



Shade the pattern on the table above
Shade and draw the next two numbers in the sequence
State the rule for the pattern, that is adding 2 to the subsequent number.

❖ Complete the arrow diagrams



❖ Given: Number sentence: 12 – 3 + 5 = ……..
Story: Twelve people got into a mini bus. At the next stop three got off and five got in. How many people are in the mini bus?

❖ Given: Story; Peter has seven brothers and 3 sisters. Altogether they are eleven.
Number sentence: 1 + 7 + 3 = 11

Given: Expression; 17 + 2
Story: Neo is 2 years older than her brother who is 17. How old is Neo?

❖ Given: Story; Thato received some sweets in a packet from his father. Tshepo got 9 sweets and Tumelo got 7 from their mother. They put their sweets together.
Expression: x + 9 + 7

Given: Story: Titoga is r years old. His mother is 25 years older than him. How old is his mother?
Expression: r + 25 years

Glossary

Mass	The amount of matter in an object (kg).
Weight	Gravitational pull on an object (Newton's).
Capacity	The amount a container can hold.
Volume	The space an object occupies.
One to one correspondence	When a mapping between two sets of the same size pairs all the elements of each set without using any element twice.
Identity element	An object in a set which, when combined (by the operation) with second object from the set, produces a result, which is equal to the second object.
Commutative	A commutative operation is one in which the order of combining the two objects does not matter.
Conservation	The concept that two equal sets of piled items remain the same, even if one is spread out.
Abacus	A counting frame that shows place value.
Solid	A three dimensional object with length, width and height.
Flat shape	A two dimensional shape with length and width.
Horizontal	Parallel to the ground (level / flat).
Vertical	Upright or perpendicular to the ground.
Pattern	Special arrangement of things.
Rectangle	A shape with four right angled corners.
Square	A rectangle with equal sides.
Semi – concrete objects	Pictures, diagrams figures and drawings are examples of semi –concrete objects.

Decomposition	Changing and carrying value from one place value to another.
Mobiles	A hanging display that can freely move, for example, hanging from a ceiling or tree.
Operation concepts	5 + 3 = 8, 5 is **addend**, 3 is also **addend** and 8 is **sum** 5 – 3 = 2, is **subtrahend**, 3 is **minuend** and 2 is **difference**
Diagonal	A line segment that joins two vertices of a shape but is not on the sides of the shape
Face	Each separate surface that makes up a solid shape
Edge	The line where two surfaces of a solid shape meet
Vertex	A point where lines or edges meet to form an angle
Associative	A binary operation which, when applied repetitively, the result does not depend on how the pairs are grouped
Cumutative	The order of combining two objects does not matter
Identity	an object in a set, which when combined by operation with any second object from the set, produces a result which is equal to the second object
Expression	a collection of quantities made up of constants and variables, linked by signs for operations and usually not including equals sign
Tessellation	an arrangement of shapes which fit together to fill a space with NO gaps or overlaps